

NANYANG TECHNOLOGICAL UNIVERSITY

School of Electrical & Electronic Engineering
Nanyang Avenue, Singapore 639798

Final Year Project

Full-Time Student Guidelines

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Compiled by

FINAL YEAR PROJECT COMMITTEE

Chairman:

Assoc Assoc/Prof Tang Xiaohong

Overall coordination, special cases, and part-time FYP

Members :

- *Assoc Prof Law Choi Look*
Calculation of staff workload and appointment of examiners
 - *Assoc Prof Xu Yan*
Scheduling of oral presentation, coordination with A/P Gooi Hoay Beng for non-graduating FYP students from Wuhan University if necessary
 - *Assoc Prof Tang Xiaohong*
Coordination of ISP, RI and SMP projects
 - *Ms. Hau Wai Ping*
Webmaster, Admin & technical support
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1. **INTRODUCTION**

As part of NTU's curriculum, all final year students of the School of EEE are required to undertake a project, supervised by one or two faculty members. This project will involve an in-depth study, investigation, construction of hardware and/or development of software and testing in any of the areas of specialized courses offered in a final year option group and spread over the whole academic year. Students are required to submit formal reports and a video file, carry out a project demonstration and also make an oral presentation on completion of the project.

The Final Year Project Committee of our School is entrusted with the task of overseeing the coordination of all the final year projects. The Committee has prepared this set of guidelines so that all students are aware of the various project requirements in terms of project schedule, project report and presentation.

1.1 **Eligibility for FYP**

The pre-requisites for EEE FYP can be found [here](#). If you are an eligible candidature, an invitation email for project selection will be sent to you from APO. All the email correspondences are to the NTU official email ids only.

1.2 **Project Duration**

Students are allocated 9 hours per week for the project, spread over two semesters. In practice, the students spend more time than this, especially during the recess weeks and the vacation period.

1.3 **Project Proposals**

Faculty members propose projects and act as supervisors. Every project proposed will contain details of various aspects of the project, including a 50-word summary of the scope, objectives etc. Projects proposed are in one of the following categories:

Type A: Internally generated project - Individual Faculty Projects

Type B: Externally generated projects - Industry Sponsored Projects (ISP), and Projects proposed by Research Institutes (RI) and other school approved organizations

The Final Year Project Committee will compile a final list of all projects, indicating the titles, name(s) of supervisor(s) and project summary, which will then be distributed to all students.

1.4 **Project Selection and Allocation**

Eligible candidates can choose to start their FYP either in Jan or Aug.

Two methods are used for the FYP project allocation. The first method is defined as **Selection-by-Supervisor** and the second method is **Allocation by Computer based on the choices selected by the students**.

1.4.1 **Selection-by-Supervisor**

Students pick their preferred projects from the list of proposed projects in the FYP web and approach the respective project supervisors for discussion. During the project selection period, if the supervisor decides to offer a project to a particular student, the main supervisor will go through StaffLink and select the allocation option to pull the project for the student. Once this has been done, the student will be allocated to that particular project and the database will then be updated to keep the other students informed. The project will only be allocated after the supervisor has completed the selection-by-supervisor process. ***Students are advised to discuss with supervisors and select their projects carefully during this phase since allocated students are not allowed to drop out after this phase.***

1.4.2 Allocation by Computer

A student can select up to 10 projects. The program allocates projects based on the order of preference from students from their 1st to 10th preferences. If a project is chosen by only one student as the first preference, it will be assigned straight away to the respective student. If there is more than one student choosing a particular project as the first preference, the program will give preference to the student who put in more choices. If all the students put in 10 choices, then a random number will be generated. This random number ranges from 1 to the total number of students choosing that project as the first choice. That particular project will be temporarily assigned to a particular student to the generated random number. After processing the first preferences of all the students, the same method is repeated for all the remaining unassigned students starting from their 2nd preference up to the 10th preference. At the end of this allocation cycle, the configuration of the project assignment and the number of the unassigned students are recorded.

A subsequent allocation cycle is repeated in exactly the same way as the previous cycle. A new project assignment together with the new number of the unassigned students are obtained and compared with the number of the unassigned students in the previous cycle. The program always keeps the configuration of project assignment that has a lower number of the unassigned students. Through processing of more than a thousand allocation cycles, the project assignment which has the minimum number of unassigned students is thus obtained. For the remaining unassigned students, a second round list containing all the remaining projects is then listed on Web and the unassigned students will be invited to select another 10 choices. Allocation is again made based on the preferences and the same allocation process is repeated. In the second round processing, all the students will be allocated a project of their choices.

1.5 Laboratory Allocation for Projects

Labs will be allocated to FYP students. Faculty members could indicate whether their project proposals have some special requests.

2. PROJECT ASSESSMENT

There are two components for the assessment of a project: 1) Interim Assessment and 2) Main Assessment which are assessed by the project supervisor and examiner. To have an independent examination in the Interim and Main Assessments, the School will appoint an examiner for each project. Both the supervisor(s) and the examiner will perform an independent evaluation.

2.1 Interim Assessment

The interim assessment is made by the supervisor(s) and examiner and is based on a Project Plan/Strategy and its implementation, an interim report as well as a video presentation about the project progress and results obtained. The supervisor and examiner will then grade the student to gauge his or her progress and performance at the end of the first semester of the project.

2.1.1 Project Plan/Strategy

During the course of the project, the students' progress will be closely monitored by the supervisor through regular meetings and/or progress reports. In the case of type B projects including ISP, SMP and RI-FYP, supervisor(s) may arrange for meetings with the students and the industry counterparts or RI co-supervisors. A short project plan/strategy report is required from the students about six weeks from the start of the project. The Project Plan/Strategy should contain the main objectives of the project, its background, the student's proposed way of carrying out the project tasks, and a proposed weekly schedule in a chart form.

2.1.2 Interim Report

It is compulsory for each student to submit an interim report at the end of the first semester. The interim report should highlight the progress made, tasks completed, results obtained, and potential problems identified. The report should be 5-10 A4 pages. The student has to submit the Interim Report to Turnitin service to check for plagiarism before submitting it to supervisor and examiner. It is compulsory for students to submit the Interim report to Turnitin service to check for plagiarism.

2.1.3 Video Presentation

Student is required to submit a video file together with the interim report at the end of the first semester. The video file should be a 2-3 minutes animated summary of the project and can contain similar content as the interim report. To learn more about video production, students can view the recording in NTULearn > FYP course site > Course Media > FYP workshop.

2.2 Main Assessment

The Main Assessment is conducted at the end of the second semester of the FYP project. It has two components: (a) assessment of Final Report and Oral Presentation, which are assessed independently by both the supervisor(s) and the examiner, (b) assessment of Demonstration by the examiner only or Final Assessment by the supervisor(s) only.

2.2.1 Project Final Report

A formal, type-written final report in one-and-a-half spacing is required from each student. The main body of the report should normally contain 40 - 60 pages. If this guideline is not complied with, the student may be asked to resubmit the report. The student has to submit the draft report to Turnitin service to check for plagiarism.

A copy of the detailed guidelines on the format of the report and other requirements for the project is attached in Appendix A.

2.2.2 Project Demonstration

Immediately after submitting the final report, it is **compulsory** for the students to make arrangements with the examiners to view a demonstration of the projects. For those projects which are research oriented or exploratory in nature, for which project demonstration may not be possible, the examiners will interview the students to gauge how well they have understood their work.

2.2.3 Final Assessment

This assessment is done by the supervisor(s) only on the student's initiative, responsibility, ability and independence, effort and contribution to project, planning, execution of plans, updating of supervisor of progress regularly, level of achievement, and outcome.

2.2.4 Oral Presentation

An oral presentation by each student is **compulsory**. During the oral presentation, each student will make a presentation. The supervisor(s) and the examiner will attend and assess the presentation. The time for an oral presentation will be 25 minutes for student, consisting of 15 minutes of presentation and 10 minutes for the question/answer session. Questions will be asked to assess the student's understanding and knowledge of the project.

Students who have valid reasons or official leave during the oral presentation period, must contact their supervisors and examiner well before the oral presentation for rescheduling.

2.3 Assessment Summary

A summary of the various assessments made by supervisor(s) and examiner is shown in the Table below.

Component	Assessor
1. Interim Assessment	Supervisor(s) and Examiner
2. Report and Final Assessment	Supervisor(s)
3. Oral Presentation	Supervisor(s) & Examiner
4. Final Report & Demonstration	Examiner

The assessment criteria for the various assessments are given in Appendix B.

2.4 FYP Schedule

Please refer to the [FYP website](#) for the various deadlines. Students should adhere to the given schedule.

3 CONFIDENTIAL REPORTS

If there is sensitive/confidential information involved, please discuss with your NTU supervisors and organisation supervisors in the company/institute (if any) on what can be included in the reports and exclude the information if necessary. Please note a copy of the FYP final report has to be submitted to the Library after the Oral Presentation. Therefore, it is important for students to clear their reports with their NTU supervisors and organisation supervisors first before submitting to Turnitin.

4. CHANGE OF PROJECT TITLE/SUMMARY/SUPERVISOR

In some cases, it may be necessary to modify/change the project title/summary. The main supervisor can do so on-line at the Stafflink from week 1 of the teaching week but changes cannot be made after the FYP committee has called for selection of projects for examination (around week 8 of the first semester of the FYP project). After the main supervisor has submitted the changes, the system will automatically send a notification email to the co-sup (if any) and the student of the affected project.

It is important for you to check that the project title, project summary, and project area are updated, if necessary, to ensure that they accurately reflect the nature and scope of the project. This is very important as examiners will be chosen/allocated based on the provided information.

5 FULL TEXT REPORT TO LIBRARY

For instruction to submit Full Text Report (softcopy) to Library through Digital Repository @ NTU, For submission instruction, please refer to <https://libguides.ntu.edu.sg/c.php?g=926884&p=6695137> . If you encounter problem with the submission, please email to library@ntu.edu.sg

6. ACADEMIC INTEGRITY

Good academic work depends on honesty and ethical behaviour. The quality of your work as a student relies on adhering to the principles of academic integrity and to the NTU Honour Code, a set of values shared by the whole university community. Truth, Trust and Justice are at the core of NTU's shared values.

As a student, it is important that you recognize your responsibilities in understanding and applying the principles of academic integrity in all the work you do at NTU. Not knowing what is involved in maintaining academic integrity does not excuse academic dishonesty. You need to actively equip yourself with strategies to avoid all forms of academic dishonesty, including plagiarism, academic fraud, collusion and cheating. If you are uncertain of the definitions of any of these terms, you should go to the [academic integrity website](#) for more information. If any material or open source codes are obtained from other sources or research team members, acknowledge it in the report. Also need to run through the Turnitin plagiarism check software to see the similarity index. Consult your supervisor(s) if you need any clarification about the requirements of academic integrity in the course.

APPENDIX A - GUIDELINES ON THE FORMAT OF THE REPORT AND OTHER REQUIREMENTS

INTERIM REPORT

An FYP interim report is a report on the current progress of the final year project. Its main role is to document what has been achieved so far and the things that need to be done later on.

Note that the FYP interim report should be 5-10 (A4) pages.

Suggested format

1. Introduction

- Provide some background information on the project, the purpose (why it is carried out), the scope of the project and the objectives of the project to help readers understand your project better.
- Include the timeline for the completion of the project (your project plan/strategy)
- State the purpose of the interim report.

2. Work Conducted

- Highlight the work that you have done so far.
- Highlight your achievement and the effort put in.
- Discuss potential problems encountered and the things that hampered the progress of the project.

3. Future Work

- Describe the work that is in progress and the work that needs to be done.
- Highlight changes made to the original requirements of the project.

4. Conclusion

- Comment on the progress of the project.
- Do an honest assessment.

FINAL REPORT

1. The main body of the report (excluding charts, diagrams, appendices, tables, references, etc.) should normally contain 40 - 60 pages. You may have to resubmit a condensed version of your report if you exceed this limit.
2. Your report should meet these typing and layout requirements:

Component	Assessor
Page	White A4 size bond paper of at least 80g /m2.
Text	Times New Roman, font size 12
Margins	35mm margin on the left hand side and 30mm on the top, bottom and right hand sides of each page.
Typing	The same font and pitch for the whole report except when highlighting important matters.
Spacing	One-and-a-half spacing.

3. To maintain consistency, all students are required to follow the same format for preparing final reports. The contents of the report should be in this order:
 - (a) Cover Page
 - (b) Table of Contents
 - (c) Abstract (*not more than one page*)
 - (d) Acknowledgements to give recognition of any advisory or financial assistance received in the course of the work on which the report is based (*optional*)
 - (e) Acronyms (*optional*)
 - (f) Symbols (*optional*)

- (g) List of Figures
- (h) List of Tables
- (i) Main chapters (Chapter 1, Chapter 2.....)
- (j) ** Discussion/Conclusion
- (k) Reflection on Learning Outcome Attainment
- (l) Acknowledging/Declaring the Use of GAI
- (m) References
- (n) Appendix (*optional*)

** Include, as many details as possible, the following topics (outcomes) and how you

- Use the techniques, skills, and modern engineering tools necessary for engineering practice with appropriate considerations for public health and safety, cultural, societal, and environmental constraints;
- Understand the impact of engineering solutions in a societal context and to be able to respond effectively to the needs for sustainable development;
- Function effectively within multi-disciplinary teams and understand the fundamental precepts of effective project management.

[Click here](#) for the report template file.

4. The Standard International System of Units (SI) should be used
5. Each reference, be it from a journal, text book or conference proceedings, should be listed consistently, as in the example below.
 - (a) Breuer, M A, and Friedman, A, Diagnosis and Reliable Design of Digital Systems, Computer Science Press, Potomac Md, 1976.
 - (b) Wakerly, J F, 'Microcomputer reliability improvement using triple-modular redundancy', Proceedings of the Institute of Electrical and Electronic Engineers, Vol. 64, No 3, March, pp 889895, 1976.
 - (c) Hata, M, Kinoshita, K, and Hirade, K, 'Evaluation of diversity effects on mobile radio system design', The Transactions of IECE of Japan, Vol. 64, No 5, May, pp 31-33, 1981.
 - (d) Comer, D J, Digital Logic and State Machine Design, 2nd ed. San Francisco, Saunders (HRW), section 7.1B, 1990.
 - (e) Mano, M M, Computer Engineering Hardware Design. New York: Prentice-Hall, sections 5.2 and 5.3, ch. 7, 1988.
 - (f) Tanenbaum, A S, Structured Computer Organization, 3rd ed. Englewood Cliffs, NJ: Prentice-Hall, section 4.2.1, 1990.
 - (g) PAL Device Data Book, Advanced Micro Devices, 1990.
6. The Final Year Project Report is an important component in the assessment of the final year project. It is written for these main readers: the supervisor(s) and the examiner for the project. The examiner is an independent party appointed by the School to assess the project. Very often the project report is the main instrument the examiner uses to judge the project. He/she may not be aware of the many hours the students may have spent on the project. The credit given to the students will be based mainly on their understanding of concepts and knowledge shown in each report. So a poorly or carelessly written report, presented after months of hard work by each student, may not be well received by an examiner.

In a project report, one generally looks for the following:

- (a) a clear idea of the scope, objectives and background of the project.
- (b) analysis of all factors in the project, noting dependency of constraints. A clear and concise presentation of any theory required must be made.
- (c) use of references which gives evidence of the students having read about the related subject. A list of references alone is insufficient. Any work, not done by the students, but included in the report, must be duly referenced. It is to be noted that plagiarism is a serious violation.
- (d) results, discussions and suggestions for further work to assess the understanding of technical concepts and the perception of the value of the work done. Results must be appropriately presented with tables, graphs, charts, etc., wherever possible and must be linked to the objectives of the project. Interpretation and discussion of results must be put into the context of the work.
- (e) a logical organization of the report to allow readers to grasp the contents readily. It is not always necessary that the information be organized chronologically. The students should put effort into thinking of the best way to present the information for the convenience of their readers. There must be a logical sequence of chapters, links between chapters, and sequence and links within chapters.

(f) a good standard of written English, proper format and layout of the report.

7. For more information on project report writing, it is suggested that the students refer to the following:

Dorothy Cheung, Lai Phooi Ching, John S T Cheung, "What Every Engineering Student Should Know About Project Report Writing," 2nd Edition, Longman Singapore, 1992

8. You need to submit a softcopy (a pdf file) of the final report to the supervisor(s) and examiner unless the supervisor(s) or the examiner requests the student to submit the final report in hardcopy (double sided, ring bound) at least 3 days in advance.
9. After your oral presentation, if the main supervisor requests, the student needs to submit a softcopy (a pdf file) or hardcopy (double sided, ring bound or hard-cover bound) of the final report or the revised final report to the main supervisor, depending on his/her.
10. For hard-cover bound, the binding should be in dark blue with gold lettering. The use of NTU logo is recommended. [Click here for the SAMPLE COVER PAGE.](#)

APPENDIX B - ASSESSMENT CRITERIA

1. Interim Assessment (by supervisor(s) and examiner)
 - 1.1 Planning and Strategy
 - 1.2 Interim Report, Results, Achievement
 - 1.3 Video Presentation
2. Main Assessment
 - 2.1 Project Final Report (by supervisor(s) and examiner)
 - 2.1.1 Presentation of the report, proper acknowledgement of other people's works
 - 2.1.2 Theory, design and implementation
 - 2.1.3 Introduction, indication of research through relevant literature review, comparison with relevant work, detailed discussion and substantiated conclusions
 - 2.1.4 Results, quality of work and level of complexity
 - 2.2 Project Demonstration/ Interview (by examiner only)
 - 2.3 Final Assessment (by supervisor(s) only)
 - 2.3.1 Initiative and responsibility
 - 2.3.2 Ability and independence
 - 2.3.3 Updating supervisor of progress regularly
 - 2.3.4 Effort, contribution to project, planning, execution of plans
 - 2.3.5 Outcome and overall level of achievement
 - 2.4 Oral Presentation (by supervisor (s) and examiner)
 - 2.4.1 Contents and organisation
 - 2.4.2 Effective presentation
 - 2.4.3 Accurate rebuttals to questions